

Model Question Paper

DIPLOMA IN COMPUTER HARDWARE ENGINEERING

Computer System Architecture

Time : 3 Hour

Max.Marks : 75

PART A

I. Answer all the following questions

(9 x 1 = 9Marks)

1	A hard disk is divided into tracks which are further subdivided into: ----A	M2.04	R
2	The _____ is the computational center of the CPU.	M3.01	R
3	A 25-pin female connector on the back of your computer will typically be ----- port	M1.04	R
4	Intel 8086 microprocessor is a ----- bit processor	M4.01	R
5	List two types of RAM	M2.04	R
6	Keyboard connected to -----port	M1.01	R
7	What are the two approaches used for generating the control signals in proper sequence?	M3.02	R
8	MIMD Stands for	M3.04	R
9	Width of flag register in 8086 microprocessor is	M4.01	R

PART B

II Answer any 8 questions from the following .Each questions carry 3 marks.

(8 x 3 = 24 Marks)

1	Compare SRAM with DRAM in terms of 1. cost 2. Power Consumption 3. Life time 4. Access time 5. Speed 6.Storage capacity 6. Usage level	M2.04	U
2	Explain functional units of a computer with block diagram	M1.02	R
3	Compare working of ink jet printer with laser printer with respect to functionality, speed , quality	M2.01	U
4	Describe any two types of display devices	M2.01	R
5	Differentiate between Von Neumann architecture with Harvard architecture	M1.02	U
6	Compare hardwired control unit design with microprogrammed control unit design	M3.02	U
7	Describe Power On Self Test (POST) function in BIOS	M1.06	R
8	Give any three components and functions of laptop computers	M1.05	R
9	Describe control flags in 8086 microprocessor	M4.01	R
10	Compare different i/o transfers	M2.02	U

PART C

Answer ALL questions. Each carries 7 marks.

(6 x 7 = 42 Marks)

III	Explain about IDE & SATA connectors used in desktop computers	M1.01	R
-----	---	-------	---

	OR			
IV	Explain any three motherboard form factors	M1.03	R	
V	Differentiate any three storage access methods OR	M2.05	U	
VI	Explain different categories of any three pointing devices	M2.01	R	
VII	Explain functions of User visible registers , Status & Control registers of processing unit OR	M3.01	R	
VIII	Explain instruction state cycle with figure	M3.01	R	
IX	Differentiate different Parallel processing techniques Flynn's classification OR	M3.04	U	
X	Differentiate between sequential and pipelined way of execution of an instruction	M3.04	U	
XI	Explain data transfer instructions of 8086 microprocessor OR	M4.04	R	
XII	Explain addition and subtraction arithmetic instructions of 8086 microprocessor	M4.04	R	
XIII	Write a pgm to add 2 , 8 bit numbers OR	M4.04	U	
XIV	Write a pgm to subtract 2, 8 bit numbers	M4.04	U	

MARKS DISTRIBUTION

Module	Hr/ Module	Marks/ Module	Type of Questions							
			PART A		PART B		PART C		Total	
			No of questions	Marks	No of questions	Marks	No of questions	Marks	No of questions	M
1	14	30	2	2	4	12	2	14	6	
2	14	30	2	2	4	12	2	14	7	
3	15	31	3	3	1	3	4	28	9	
4	15	32	2	2	1	3	4	28	7	
Total	58	123	9	9	10	30	12	84	29	1

Cognitive level Marks distribution

Cognitive level	Marks	%of Marks
Remembering	73	60
Understanding	50	40
Applying		
Total	123	100